



Maize Production for Food, Feed and Fodder



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डॉ. तिलक राज शर्मा

उप महानिदेशक (कसल विज्ञान)

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Maize (*Zea mays* L.), the queen of cereals, surpasses all other food crops in its ability to adapt to the diverse agro-ecological niches and being cultivated from 58°N to 55°S latitude. It is the leading cereal crop in the world with more than one billion tonne production. In India, it holds third rank among the cereals, after rice and wheat in terms of area and production. Traditionally, maize is grown as a *kharif* crop in India, but from 1970s its expansion started in *rabi* season and later on in the *spring* season too. Now, the crop is grown round the year and in all agro-ecologies throughout India, not only for grain but also for specialized purposes like sweet corn, baby corn and for silage making.

Maize is widely used as food, feed, fodder, industrial raw material and in recent past for bio-fuel production. It is a key crop in doubling farmers' income as maize can be integrated effectively with dairy, poultry, fisheries and piggery. Speciality corns has a special role in value chain development. Considering the changing food habits in India, the demand of maize is supposed to increase in the near future. To meet the growing demand, there is a need to enhance maize productivity through adoption of the best production practices by the maize farmers.

I am happy to know that ICAR-Indian Institute of Maize Research, Ludhiana is bringing out a publication on "*Maize production for food, feed and fodder*" which have complete cultivation information for normal and specialty maize right from the sowing to harvesting. I congratulate the authors for their contributions in bringing out this comprehensive and important publication. I am sure that this publication will be useful for researcher, farmers and other stakeholders for enhancing the maize production and productivity in the country.

(T R Sharma)

PREFACE

Declining water resources, degrading soil health, higher production cost, lesser profitability and climate change are the major challenge before Indian agriculture. Resource use smart crops which can give more profit from lesser resources are need of the hour. Maize is one such crop with highest production potential and comparatively lesser water foot-prints. It is a multi-purpose crop being used as food, feed, fodder, and raw material for various industrial products including bio-fuels.

Maize is the leading cereal crop in the world with more than one billion tonne production and in India it holds third rank after rice and wheat in terms of area and production. Traditionally maize in India used to be consumed as food crop. However, over time maize has become rather an industrial crop than food crop as over 75% of the maize produced in India is used for industrial purchases, mostly as feed and source of starches. Being industrial in nature maize can ensure higher remuneration. Further, specialty corns, baby corn and sweet corn with short to medium duration cropping period can play important role in diversification in peri-urban agricultural systems and nutritional garden. They fit well in integrated farming system, where the byproduct of specialty corn, i.e. green fodder can support livestock very well. During last 50 years, maize productivity has increased from around 0.99 t/ha to 3.1 t/ha but still we are far behind leading maize producing nations. Lesser adoption of improved production practices is one among the key factors contributing low productivity of Indian maize, besides prevalence of rainfed growing condition, limiting effects of various biotic and abiotic stresses among others. Improved high yielding cultivars coupled with better crop management is bound to increase productivity of maize in India.

The compiled production practices of various types of maize is expected help to researcher, extension workers and farmers in enhancing maize production and productivity in the country.

Authors

April 2020
Ludhiana (Punjab)

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ICAR

Introduction

Climate change has far serious implications concerning the food and nutritional security of nations like India where agriculture is the backbone of nation's economy and growing population is the biggest reason to worry. Rural India largely depends on its natural resources and farming for earning its livelihood and food security and climate change affects the natural ecosystem progressively. In future, it will be even more difficult to predict occurrences of phenomena such as droughts, floods, cloud bursts, etc. due to climate change. Consequently, farmers must adapt to the changing climate to ensure optimum crop yields and farm income. Farmers in general and small and marginal farmers in particular must enhance the resilience of agriculture to face the upcoming challenges. Transforming agriculture through adoption of climate resilient practices and technologies would be inevitable to stabilize agricultural production and enhance farmer's income.

Participatory demonstration of location specific and climate smart technologies is necessary for enabling farmers to cope climatic variability and extreme weather phenomenon. Adoption and spread of these climate resilient technologies would help farmers fetch adaption gains and reduce the Green House Gas (GHG) emissions alongside. Considering the urgent need to address the issue of climate change at the farm level, Indian Council of Agricultural Research (ICAR) launched National Initiatives in Climate Resilient Agriculture (NICRA) in February, 2011, which was renamed as National Innovations in Climate Resilient Agriculture (NICRA) to emphasize the role of evolving innovations. Resilience is the capability of the production system to resist negative impacts of climate change and also the capacity to recover quickly after the damage. Therefore, National Innovations in Climate Resilient Agriculture (NICRA) was formulated to develop and demonstrate region specific improved technologies that would enhance the resilience of Indian agriculture to climate change thereby addressing climate vulnerability and its negative impacts. The emphasis on adaptation to climate variability necessitates appropriate responses to contingency situations. Considering the need to enhance the resilience of Indian agriculture production system to climate variability and climate change, the center theme should be the Sustainability of the production systems facing natural resource degradation. Thus, the centre of attraction in NICRA is not the enhancement of productivity but the ability of the existing system to cope with the vulnerability and to improve the natural resource use efficiency for sustaining the productivity gains that have been achieved already. The project aimed to enhance resilience of Indian agriculture to climate change and climate variability through its different components namely Strategic Research, Technology Demonstration, Capacity Building and Sponsored/Competitive Grants.

Uses of Maize

Feed

Approximately 60% of total maize production in India is used for preparation of concentrated feeds for cattle, poultry and piggery as source of energy. Fishery feed also uses maize as an important component. The yellow colour of maize is very useful in providing yellow colour to egg yolk and yellow tinge to the milk.

Food

Grain maize, pop corn, baby corn, sweet corn and green maize cobs are widely used as human food. Most commonly used forms are as Chapattis, porridges, corn flakes, boiled or roasted green ears, pop corn, baby corn products like salad, vegetable, pickle, pakoda, kheer, laddu and halwa; sweet corn products like soup, chat, corn cheese balls, masala corn and salad etc. Around 13% of total maize grain production in India is used as food and 7% as processed food, however this does not include use of specialty maize, particularly baby corn, sweet corn and green cobs.

Industrial uses

Maize acts as a raw material in the manufacture of starch, syrup, dextrose, oil, gelatine etc. Corn flour is used as a thickening agent in the preparation of many edibles like soups, sauces and custard powder. Corn syrup is used as sweetener in processed foods such as soft drinks and candies. Corn sugar (dextrose) is used in pharmaceutical formulations. Maize oil is widely used as a cooking medium and it has the quality of reducing cholesterol in the human blood. Maize oil has use in cosmetics and pain industry as well. Corn gel because of its moisture retention character is used as a bonding agent for ice-cream cones. Maize stalks are made into paper and wall board; husks are used as filling material; cobs are used directly for fuel, to make furfural, fermentable sugars, solvents, liquid fuels, charcoal, pulp, paper and hard boards. Approximate 14% of total maize production in India is utilised for industrial purposes.

Bio fuel and bioplastic

Maize is also used for liquid bio-fuel (ethanol) production in USA. The ethanol is used as transportation fuel after blending with gasoline. India is also slowly opening up use of maize grain for ethanol production. The maize starch is also used for making biodegradable plastic using polylactic acid (PLA) which will be very useful in the scenario of banning single use of plastic.

Forage and silage

Maize stalk is used to feed dairy animals in the form of green fodder, dry roughages and silage. Maize fodder has higher digestibility than any other non-leguminous forage crops and it does not contain any toxic substances. Maize is also preferred for silage making over other fodders, as maize plant is easy to chaff, requires less labour and its silage is soft.

Brief Cultivation Facts

A. Maize for grain

A.1. Irrigated kharif maize

Sowing time: Generally, the ideal sowing time is June 20 to end of July. However, in fields where water stagnation may occur, early sowing is desirable so that plant reaches a firm stand to avoid lodging associated with water logging.

Soil: Maize is very sensitive to water-logging and considerable yield losses occur if, the crop faces water-stagnation for more than two days. Hence, it is better to plant the crop on well-drained sandy-loam to silty-loam soils.

Land preparation: Maize needs well pulverized and smooth field for quick seed emergence and root growth. Hence, two harrowing followed by one planking is needed for field preparation and early season weed management. Maize can also be successfully produced under zero-till conditions using happy seeder or zero till multi-crop planter.



Fig: Land preparation (a) harrowing; (b) land leveling

Farm-equipments: Mould board plough (MB plough), disc or tine harrow, land-leveler, maize planter, narrow bed planter, pneumatic planter, wide bed planter etc.

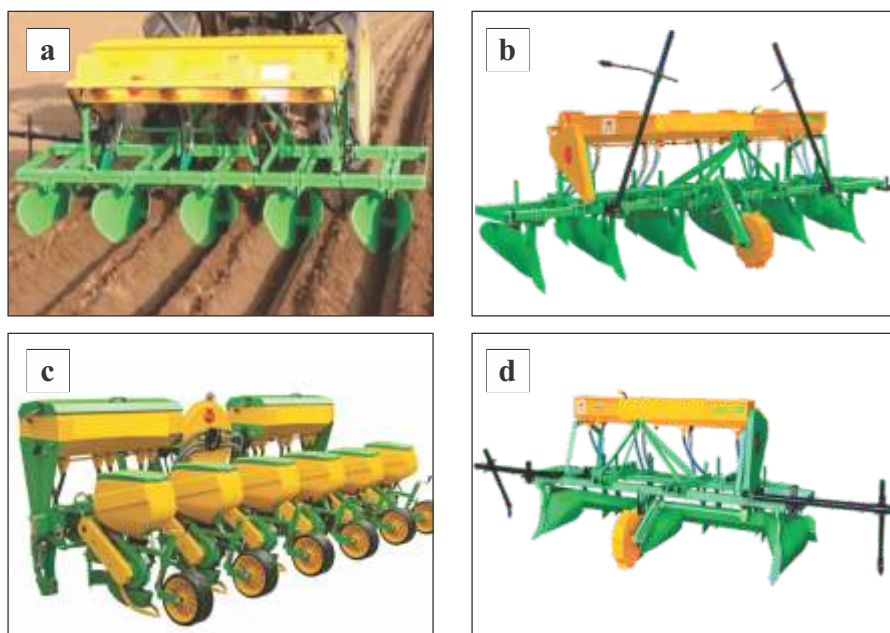


Fig: Maize planting equipment (a) Ridge planter; (b) Maize planter; (c) Pneumatic planter; (d) Wide bed planter

Seed rate and seed treatment: Eight kg seed of field corn is needed for seeding in one acre area. Seeds should be treated with any of the following fungicides and insecticides before sowing to protect the crop from most prevalent (seed and soil borne) diseases and insect-pests.

- 1) Bavistin + Captan in 1:1 ratio @ 2g/kg seed for Turicum leaf blight, Banded leaf and sheath blight, Maydis leaf blight etc.
- 2) Apran 35 SD @ 4g/kg seed for Brown stripe downy mildew
- 3) Captan 2.5g/kg for Pythium Stalk Rot
- 4) Imidachlorpid (Gaucho) @ 6 ml/kg or Fipronil @ 4ml/Kg seed for termite and shoot fly.
- 5) Thiamethoxam + Cyantraniliprole @ 4 ml/kg seed for fall armyworm.

Any of the above seed treatment may be selected depending upon predominant biotic stresses.

Sowing method: In *kharif* sown maize, raised bed planting is recommended to protect the crop from water logging. In raised bed planting, 70 cm wide ridges (40 cm ridge and 30 cm furrow) are prepared with the help of bed planter. Bed planter with incline plate seed metering system can precisely place the maize seed at required depth and does simultaneous operation of raised beds making and planting. One line of maize on each raised bed is desirable when sole crop of maize is planted keeping seed to seed spacing at 20 cm. Optimum plant density (30,000/acre) should be maintained to tap full potentials of hybrids. Southern side planting is advised on East-West oriented

ridges. Happy seeder or zero-till bed planters with inclined plate to be used for crop establishment under zero tillage or for sowing under crop residue.

Sowing depth: For proper germination and early vigor, seed should be sown at 3.5-5.0 cm depth.

Inter cropping: One row of either (a) Cowpea for fodder; (b) Soybean/urdbean/mungbean for grain; (c) Groundnut for pods, in between two consecutive rows of maize is desirable for inter-cropping with maize. The shade loving crops like turmeric and ginger can also be successfully cultivated as intercrop with maize. Herbicides should not be used for weed control in inter-cropping system. In case of intercropped maize slightly higher doses of fertilizers should be applied than sole crop.

Irrigation management: Water requirement of maize is 500–800 mm per growing season. During *kharif* season, partially water requirement is met by rainfall. Hence, 1-4 irrigations are required, which may vary depending on frequency of rainfall. Adequate moisture at germination, pre-tasseling, silking and grain-filling stages should be ensured through irrigation, if rainfall is not there. If available, the sprinkler irrigation up to knee length stage is very good for maize crop. In ridge-sown crop, irrigation should be given in furrow up to 2/3 of height of the ridge.

Nutrient management: Hybrid maize is very responsive to nutrient application and has slightly high nutrient requirement as compare to other cereals due to high yield potential. It can be grown both organically or by integrated nutrient management involving organic and inorganic nutrient supplementation.

Organic nutrient management: Following options for nutrient management in organic maize production to be used based on availability in suitable combination

- Green manuring: Green manuring crops like dhaincha/sunhemp/cowpea at 12/20/20 kg seed rate/acre, respectively are very useful. Fifty days old crop is to be ploughed down and reserve the field for 10 days for decomposition before sowing of maize.
- Straw of summer mungbean/cowpea may be buried before sowing of maize
- Farm yard manure/Compost @ 6 tonnes/acre or vermicompost @ 3 tonnes/acre
- Application of Azatobacter/Azospirillum with PSB and NPK consortia for seed treatment @ 200 g/acre or liquid formulation @ 100 ml/acre.

Integrated nutrient management: Application of 6 tonnes/acre well decomposed farm yard manure/compost mixed with Azatobacter/Azospirillum with PSB, VAM and NPK consortia @ 5-6 kg/acre or seed treatment with Azatobacter/Azospirillum with PSB and NPK consortia @ 200 g each/acre or liquid formulation @ 100 ml/acre needed for better moisture retention and initial boost of the crop. Macro- and micro-nutrient requirements of the crop need to be accomplished through use of suitable fertilizers. The following schedule of fertilizer may be used:

Crop Stage	Fertilizers scheduling
Sowing (Basal)	Drill full dose of P, K and 1/3 of N-fertilizer dose, micro-nutrient application
Knee-high (1 st split)	Top dress 1/3 N-fertilizer dose and micronutrient spray
Pre-tasseling (2 nd split)	Top dressing of remaining 1/3 N-fertilizer dose and micro-nutrient spray

Rate of different fertilizers

Varieties	Schedule	Fertilizer (kg/acre)			
		Urea	DAP	MOP	ZnSO ₄ .7H ₂ O
Long duration hybrid	Basal	23	52	27	10
	1 st split	44	--	--	--
	2 nd split	44	--	--	--
Short duration hybrid, composite	Basal	14	35	20	10
	1 st split	26	--	--	--
	2 nd split	26	--	--	--

Maize is sensitive to Zinc deficiency. Zinc deficient crop shows stunted growth and develop short inter-nodes. A white (or pale-yellowish) tissue with reddish veins appears on leaf blade. To mitigate it 10 kg/acre ZnSO₄.7H₂O or 6.5 kg/acre ZnSO₄.H₂O should be applied at sowing. Apply 10-15 kg Sulphur and 0.5 kg Boron/acre in deficient soils.



Fig: Zinc deficiency symptoms in maize

Weed management: Weeds significantly affect maize production and the critical period for crop-weed completion is 15-45 days after sowing.

Cultural control of weeds: a) Two manual hoeing at 15 and 30 DAS; b) Inter-cropping of one or two rows of fodder cowpea in between maize rows reduces weed problem considerably; c) Always use weed free seeds for sowing; d) Do not spread un-decomposed compost in field because it may contain viable weed seeds; e) Do not allow the seed formation of weeds in fields.

Chemical control of weeds: Pre-emergence Atrazine application @ 400 g ai/acre followed by one hand weeding at 35-40 days gives good weed control. Recommended dose of Atrazine at pre-emergence followed by either of the post-emergence herbicide (mentioned below) for effective weed control is also equally effective.

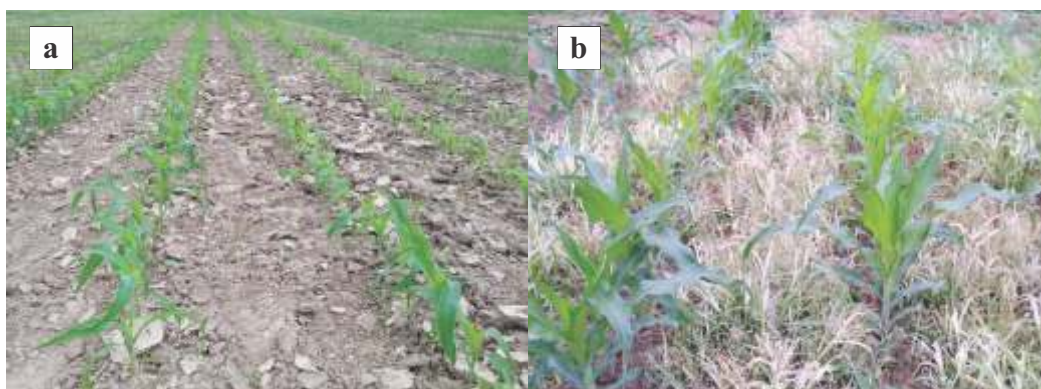


Fig: (a) Maize field after weeding; (b) Maize field after Tembotrione spray

Selective herbicides for maize

Herbicide	Formulations	Dose (g a.i./acre)	Time of application (DAS)
Atrazine	50% WP	400	0-2 (pre-emergence)
Topramezone	33.6% SC	10	20-30 (Post emergence)
Tembotrione	34.4% SC	48	20-30 (Post emergence)

Precautions:

- Use flat fan nozzle for herbicide spray.
- Do not use under/over dose of herbicide.
- While spraying weedicide sprayed field should not be stepped on, i.e. the farmer should move backward or sideways.

Saving crop in case of excess water damage: If water logging occurs in the field and damage is moderate, 6 kg/acre of urea in two sprays at weekly intervals may be applied. In case of moderate to severe damage, 25-50 kg urea/acre after the water flooding is over may be broadcasted.

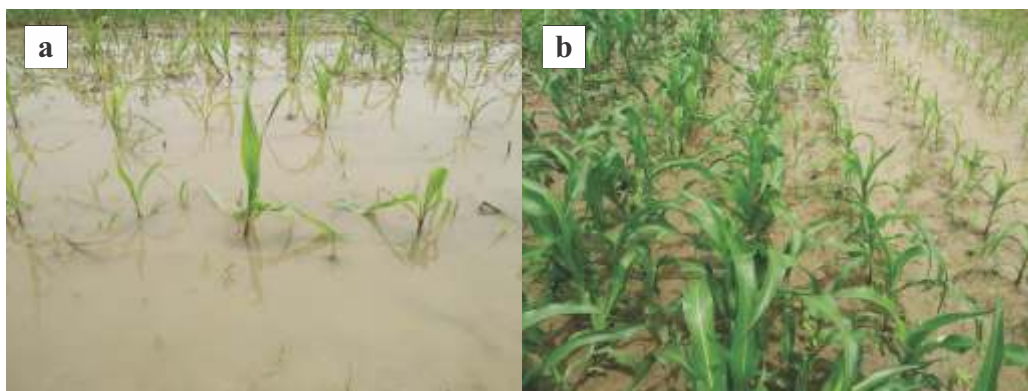


Fig: (a) Water-logged maize field (b) Growth reduction due to water-logging

Plant Protection

Major diseases: Maydis Leaf Blight (MLB), Bacterial Leaf and Sheath Blight (BLSB), Charcoal Rot (CR), Bacterial Stalk Rot (BSR), Rajasthan Downey Mildew (RDM), Curvularia Leaf Spot (CLS), Post Flowering Stalk Rot (PFSR), Brown strip Downey Mildew (BSDM) and Common Rust are major diseases of maize.

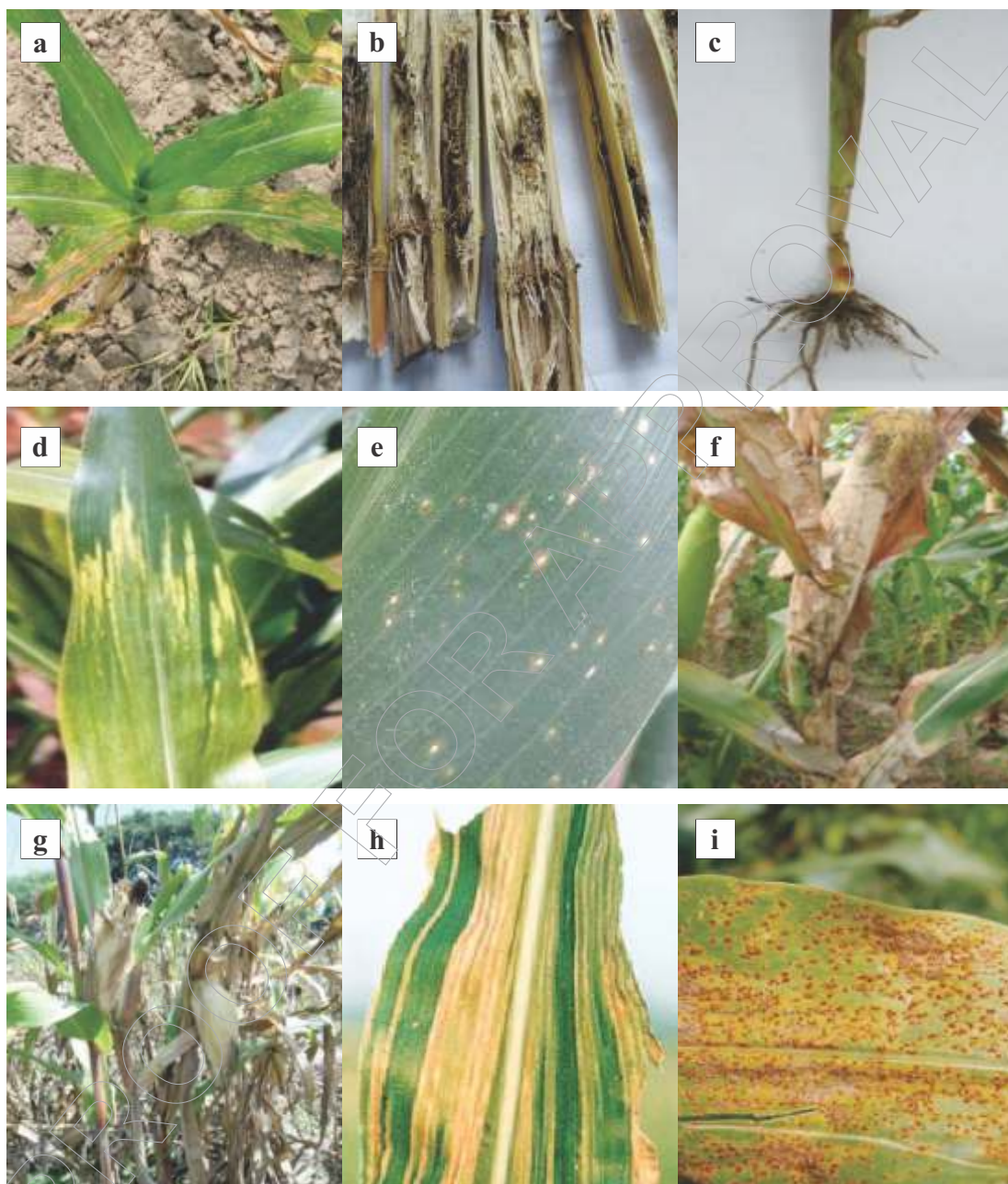


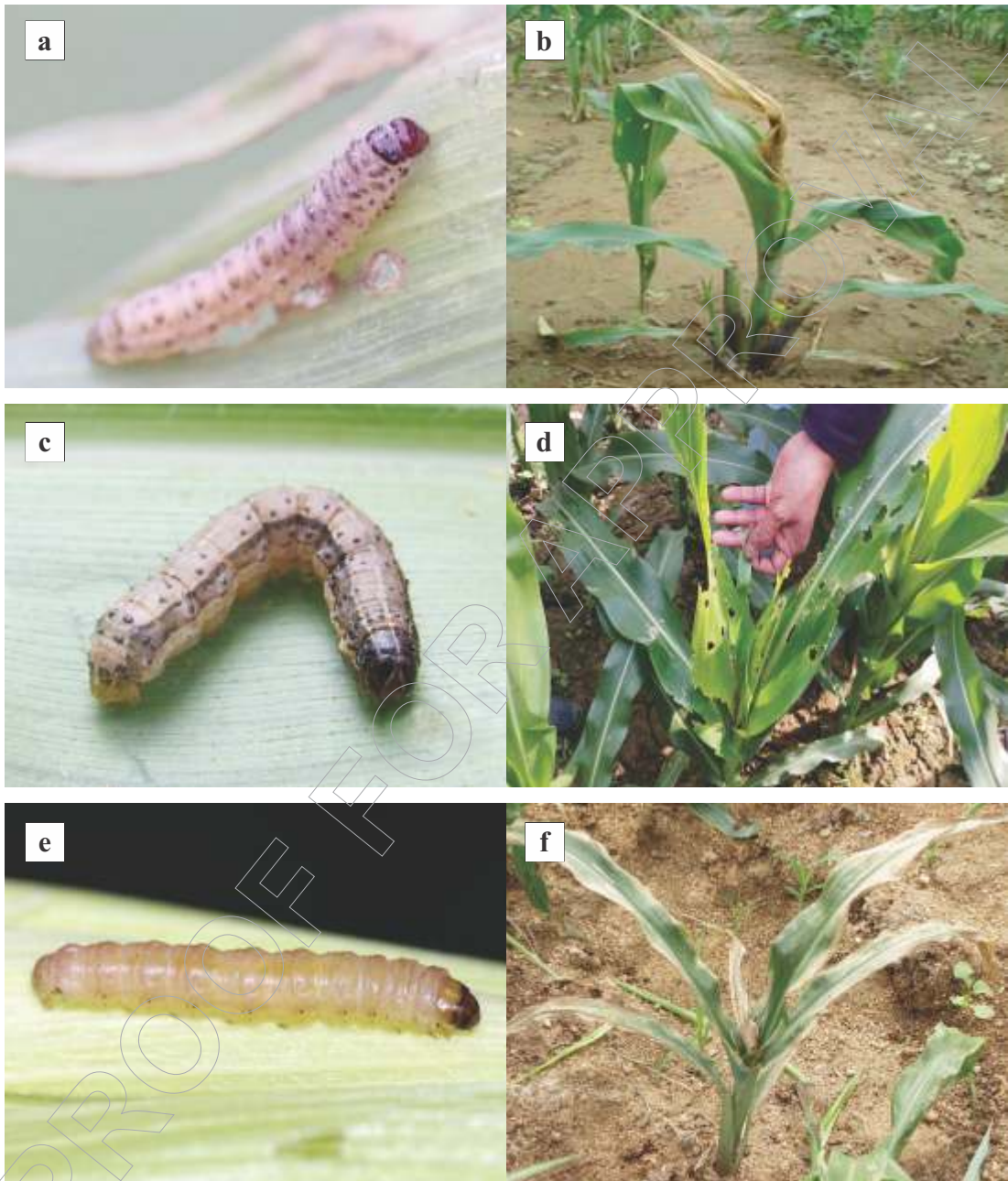
Fig: Maize diseases (a) Maydis leaf blight; (b) Charcoal Rot; (c) Bacterial stalk rot; (d) Rajasthan downey mildew; (e) Curvularia leaf spot; (f) Bacterial leaf and sheath blight ; (g) Post flowering stalk rot; (h) Brown strip downey mildew; (i) Common rust

Disease management in maize

Disease	Cultural control	Chemical control
Maydis leaf blight	Deep ploughing of previous crop residue through	A) Seed treatment with Bavistin + Captan in 1:1 ratio @ 2.0 g/kg seed B) Spray Dithan Z-78 or Zineb @ 2.4-4 g/litre water and repeat the spray at 8-10 days interval C) Grow MLB resistance varieties, if the disease incidence is regular
Bacterial leaf and sheath blight Charcoal rot	Stripping of two lower leaves along with leaf sheath A). Deep ploughing, B). Crop rotation C). Maintain proper moisture during flowering	A) Seed treatment with Bavistin + Captan in 1:1 ratio @ 2.0 g/kg seed B) Spray of Sheethmar (Validamycin) @ 2.7 ml/litre water
Bacterial stalk rot Rajasthan downey mildew	Stripping of two lower leaves along with leaf sheath A). Field sanitation B). Destroy infected plant	A) Seed treatment with metalaxyl (Ridomil 25 WP, Apron 35 SD) @ 2.5 g/kg B) Foliar spray of metalaxyly (Apron 35 FN) @ 2-2.5 g/liter of water at very first appearance of disease
Curvularia leaf spot		A) Spray (Carbendazim 12% + mecozeb 62.7%) @ 2 g/liter water 35 and 55 DAS B) If zineb is available, then spray Zineb 75% @ 2 g/liter at 35 and 55 DAS

Disease	Cultural control	Chemical control
Post flowering stalk rot	A) Maintain optimum plant population B) Avoid excess N C) Follow crop rotation with non-host crop like soybean D) Manage attack of stem borer as their injury predisposes to stalk rot	- - -
Brown strip downey mildew	A) Field sanitation B) Destroy infected plants C) Weed control D) Maintain optimum plant stand E) Planting before rainy season F) Maintain low seed moisture (9%) at planting	A) Seed treatment with metalaxyl (Ridomil 25 WP, Apron 35 SD) @ 2.5 g/kg B) Foliar spray of metalaxyly (Apron 35 FN) @ 2-2.5 g/litre of water at very first appearance of disease
Common rust	A) Field sanitation B) Destroy infected plants C) Weed control D) Maintain optimum plant stand E) Planting before rainy season F) Maintain low seed moisture (9%) at planting	Spray Diathane M-45 @ 2.4-4 g/litre water and repeat the spray at 8-10 days interval

Major Insect Pests: Maize stem borer, Pink stem borer, Shoot fly and Fall army worm are major pest of maize Among these pests, Maize stem borer occurs in *kharif*, Pink stem borer in *rabi* and Shoot fly in *spring* season.



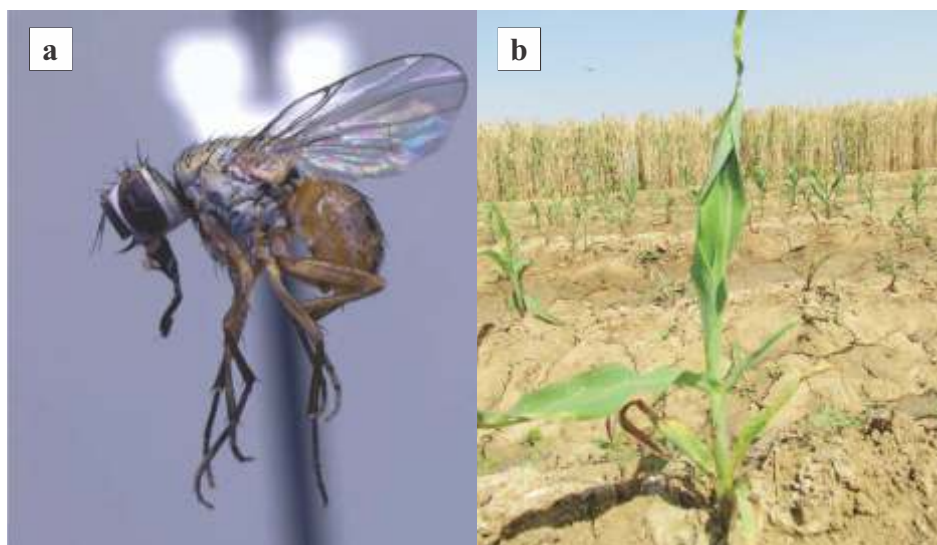


Fig: Maize pests (a) Maize stem borer; (b) Damage symptoms Maize stem borer; (c) Fall armyworm larvae; (d) Damage symptoms of FAW; (e) Pink stem borer; (f) Damage symptoms of Pink stem borer; (g)Shoot fly; (h) Damage symptoms of Shoot fly

Management of maize pests

Pest	Cultural control	Chemical control	Biological control
Maize Stem Borer (Kharif season pest) and Pink stem borer (rabi season pest)	Grow cowpea in between maize rows as intercrop against Maize stem borer. Border planting of sorghum (3-4 rows) around maize crop against Maize stem borer.	Spray the crop with chemicals like Chlorantraniliprole 18.5% SC @ 60 ml per acre at 15-18 DAS Place a pinch of Carbofuran 3% G in whorl of infested maize plant at initial stage of infestation	Use Trichocards (Trichogramma chilonis) on 15 days old crop @ 3 cards per acre

Pest	Cultural control	Chemical control	Biological control
Fall Army Worm	Choose cultivars with tough husk cover Border planting of sorghum (3-4 rows) around maize crop against Maize stem borer. Intercropping with pulses and other non-host crop	Whorl application of either of pesticide A) Spinetoram 11.7% SC @ 0.5ml/l B) Thiamethoxam 12.5% + Lambda Cyhalothrin 9.5% ZC @ 0.25 ml/l C) Chlorantraniliprole 18.5% SC @ 0.4 ml/l D) Emamectin benzoate 5 SG @ 0.4 g/l --	Whorl application of either of biological formulations at 15-25 days after sowing A) 5% NSKE or azadirachtin 1500 ppm @ 5 ml/litre B) Metarhizium anisopliae talc formulation (1x10⁸ cfu/g) @ 5g/litre C) Nomuraea rileyi rice grain formulation (1x10⁸ cfu/g) @ 3g/litre D) Btk formulation @ 2ml/l of water Release of Trichogramma pretiosum @ 50000 Or Telenomus remus @ 10,000 per acre at weekly intervals
Shoot fly (spring)	Early planting in first week of February	A) Seed treatment with Imidacloprid 600 FS (Gaucho) @ 6 ml / kg or Thiomethoxam 30 FS @ 8ml/kg seed. B) Soil application of Furadan (Carbofuran 3G) @ 8-10 kg per acre in the furrow during sowing is also helpful agains	--

Application of nitrogen and irrigation after control measures helps in faster recovery from pest damage.

Bird management: Nearly ten bird species feed and damage maize cobs. The yield loss due to bird feeding is huge and ranges between 10 to 40% in the case of maize crop. To reduce the damage by Parakeets and crows, cover maize cobs by wrapping adjacent green leaves around them on outer four rows of the field. Acoustic devices also available in market which keep away depredatory birds from fields by producing recorded sounds. These devices produce natural sound of bird predators and alarm call of pest birds. Thus, the pest birds avoid the broadcast area. Beside this, reflective ribbons also scare birds through reflection of sunlight and humming noise produced by the wind.

Harvesting and shelling: Maize can be harvested when the husk has dried and turned brown. Apart from hand-picking, combined harvester may also be used for quick harvesting. After manual harvesting, depending upon the cultivated area, manual shellers or maize dehusker-cum-sheller or maize thresher may be used for separating grains from cobs.



Fig: Combine harvesting in maize

Grain drying: During harvesting of maize, grain moisture content is quite high (~30-35%). Due to higher moisture content farmers cannot get good price of their produce. Beside this, storage at high grain moisture creates problems of fungal infection and can cause heating and loss of germination. Hence, after crop harvesting, produce drying is very much essential. Portable Maize Dryer can be used for reducing the moisture content of the cobs. It is tractor or electricity driven, portable and long lasting dryer. It can dry any type of grain and do not need pre-cleaning of grains. Its drying rate ranges between 2-10 t/h depending on crop type, grain moisture content etc. The optimum moisture in grain for long-term storage should be below 14%. Sun drying of the cobs and seeds is required in absence of other drying options prior to marketing as moisture above optimum level reduces market prices and increases chance of aflatoxin contamination.



Fig: Mobile batch dryer

Grain storage: Large scale *kharif* maize harvesting during October-November and *rabi* maize in April-May months, result in price drop. Hence, to fetch better price, farmers generally store their produce for higher profitability. But, the higher atmospheric humidity, cloudy weather and higher grain moisture content leads to fungal infection and physical deterioration of maize grain. Hence, proper drying followed by moisture proof storage are very much needed for long term storage of maize grains. Pusa bin and metal bin are some cost-effective and moisture proof storage structures which can be used by maize farmers. Pusa bin is modification of ordinary mud storage structure. It is rectangular in shape and constructed of bricks and earth. To make it moisture proof, a plastic film (700 gauge) is used all around ((inner side) of bin. It can store up to 3 tonnes of well dried grains for longer period with minimum loss. Metal bin is moisture proof storage structure and made of steel or aluminium. The bin is durable and also commercially available. Its storage capacity ranges from 1 to 10 tonnes. Hermatic cocoons are another storage container having two plastic half-joined together with a air tight zip. It is made up of thin and ultra low permeable plastic with 500 times lesser permeability to oxygen than normal plastic. The air tight container prevent development of storage pest as it block respiration of pest (due to low oxygen). It does not allow moisture movement hence also prevent aflatoxin problem.

Store grain pest management: Rice weevil and Angoumois grain moth are major pest of maize during storage. Following point should keep in mind during storage.

- The moisture content of grain should be less than 12%.
- Clean the storage bins before using them for storage.
- Newer grains should not be mixed with older ones.
- Staggered sun drying with short exposure to sun spread reduces insect infestation.

- Use of plant products such as *Adathoda vasica*, *Azadirachta indica*, *Vitex negundo*, *Catharanthus roseus* @ 2% w/w (20g /kg seed) have been found to be effective against storage pests.

A.2. Rainfed kharif maize

Sowing time: In rainfed regions sowing time is rain dependent and should be done just after monsoon occurrence in end of June to second fortnight of July.

Seed rate and spacing requirement in rainfed maize

Seed rate (kg/acre)	Spacing (cm)	Plant population (per acre)
7-8	70 x 20 -25	23,134 to 28,918

Fertilizer scheduling: Organic nutrient supplementation to be given and inorganic N fertilizers is to be splitted into two doses, at basal and depending on moisture availability between knee high to tasseling stage.

Rate of different fertilizers needed for rainfed maize crop

Soil Type	Schedule	Fertilizer (kg/acre)		
		Urea	DAP	ZnSO ₄ .7H ₂ O
Sandy loam to clay loam soil with adequate moisture stored	Basal 1 st split	46	35	10
		24	--	--
Loamy sand to sandy soils with low moisture stored	Basal 1 st split	35	18	--
		--	--	--

Soil moisture conservation: For rainfed crops, soil moisture is the most limiting factor. Hence, moisture conservation practices are required to be followed to attain maximum yield.

Soil moisture conservation practice.

- Plough the field against slope after pre-monsoon showers to enhance water absorption/filtration
- Sowing and other operations should be carried out on contour/across the slope
- Spread locally available mulching material in the last week of August
- Application of the farm yard manure/compost @ 5-6 tonnes/acre

Rest agronomic practices are same as discussed earlier.

A.3. Rabi maize

Sowing time: 15th October to 15th November is the best time, however this may vary for region to region.

Sowing method: Before sowing, seed should be soaked overnight in warm (45°C at the time of seed soaking) water. This treatment helps in obtaining better plant stand and healthy crop. Sowing should be done on the southern side of the east-west ridge so that the optimum amount of sunshine is received and the seedbed remains warm.

Seed rate and spacing requirement in rabi maize

Seed rate (kg/acre)	Spacing (cm)	Plant population (per acre)
20-22	60 x 20	33,333

Intercropping: During the *rabi* season potato, table pea, onion, garlic, coriander, spinach, coriander, beetroot, cabbage etc. can be grown in between maize rows to ensure higher income of growers without harming the yield of winter maize.

Fertilizer scheduling: The following fertilizer schedule should be used to get higher yield.

Crop stage	Schedule	Fertilizer (kg/acre)			
		Urea	DAP	MOP	ZnSO ₄ .7H ₂ O
Before at sowing	Basal	26	70	40	10
4 leaf stage	1 st split	26	-	-	-
8 leaf stage	2 nd split	40	-	-	-
Tasseling stage	3 rd split	26	-	-	-
Grain filling stage	4 th split	6	-	-	-

Irrigation management: Four to six irrigations are needed for *rabi* season crop. If six irrigations are given, they should be applied as two before flowering, one at the time of flowering, two after flowering and one at the early grain-filling stage. If only five irrigations are given, one irrigation at the vegetative stage may be avoided; and if only four irrigations are given, irrigation after the dough stage may be avoided. The irrigation should, however be changed suitably if adequate rains are received. Rest management practices are similar to irrigated maize.

A.4. Spring maize

Sowing time: January 20 – February 20 is ideal sowing time for *spring* maize planting.

Sowing method: Evaporative losses of water during *spring* season from the soil under flat as well as raised bed planting is higher and hence crop suffers from moisture stress. Under such conditions, it is always advisable to grow maize in furrows for proper growth, seed setting and higher productivity. Zero-till planning is beneficial in *spring* maize after harvesting of potato, mustard etc.

Irrigation: *Spring* maize needs nearly 10-15 irrigations sometimes even higher as the evapo-transpiration demand is quite higher in summer season. Hence, a water use-efficient irrigation system i.e. drip method is desirable under *spring* sown maize. Apply first irrigation at 25-30 DAS. Under non-drip irrigation, apply irrigation at 2 weeks interval upto 10th April and on per week basis after this, up to maturity. From water conservation point of view *spring* maize without drip irrigation is not recommended.

Fertilizer scheduling: For higher yield, integrated nutrient management found beneficial for *spring* maize and organic sources option mentioned in *kharif* maize to be used along with chemical fertilizer mentioned as below:

Rate of different fertilizers needed

Varieties	Schedule	Fertilizer (kg/acre)			
		Urea	DAP	MOP	ZnSO ₄ .7H ₂ O
Medium duration hybrid	Basal	33	55	27	10
	1 st split	33	--	--	
	2 nd split	33	--	--	
Medium duration hybrid	Basal	25	27	27	
	1 st split	25	--	--	
	2 nd split	25	--	--	

Rest management practices are similar to irrigated *kharif* maize for grain purpose.

B. Specialty corn/special purpose maize

B.1. Pop corn

Pop corn cultivation practices are similar to grain purpose as discussed earlier except the pop corn cultivars are shorter of duration (80-90 days). However, as a premier crop preferably to be grown under irrigated conditions. This crop needs isolation of at least 400 m from normal maize as pollination with normal maize deteriorates its popping quality. Generally, 10-15 days isolation in sowing time also ensures quality pop corn. As the plant type of the pop corn is weak and affected more by diseases and pest and thus should be preferably to be grown in mild climate i.e. winter season/hilly areas for better yield and quality.

Seed and spacing requirement

Seed rate (kg/acre)	Spacing (cm)	Plant population (per acre)
4 - 5	60 x 20	33,724

Rest management practices are similar to irrigated maize for grain purpose.

B.2. Baby corn

It is shortest duration maize crop and gets ready for harvesting within two months of planting and harvested in 60-70 days during *kharif* and 80-100 days in *rabi* season. This crop needs isolation of at least 400 m from normal maize as pollination deteriorates its quality. Generally, 10-15 days isolation in sowing time also ensures quality baby corn. Detasseling is needed in baby corn for quality assurance. It is done by removing the tassel of the plant as soon as it emerges from the flag leaf. It should be practiced row-wise. While detasseling, leaf should not be removed which will otherwise affect net photosynthesis and ultimately reduce average baby corn yield. However, this is not needed when male sterile baby corn cultivars are chosen for cultivation. Harvesting should be done in morning or evening and stored under cool conditions in shed. After dehusking cobs should preferably kept under wrap and refrigerated. The crop should be harvested after 1-3 days of silk emergence. The remaining plant portion (tassels, main stem, husks) serves as nutritious green fodder for livestock.

Sowing time: The sowing to be avoided in areas receiving temperature of less than 10°C at any of the crop growth stages for higher yield of baby corn. It can be sown round the year in India expect second fortnight of December to end of January in Northern India.

Seed and spacing requirement

Seed rate (kg/acre)	Spacing (cm)	Plant population (per acre)
10 - 12	60*15-20	33,724 to 44,966

Intercropping: In general, short duration varieties of intercrops to be preferred for intercropping with baby corn. In *kharif* season, cowpea for green pods and fodder purposes and coriander for green leaves can be intercropped with baby corn. Baby corn can be intercropped with vegetable crops, viz., spinach, cabbage, cauliflower, coriander, sugar beet, radish etc which is particularly suitable in *rabi* season baby corn. Recommended dose of fertilizers of intercrops should be applied in addition to the recommended dose of fertilizers of baby corn.

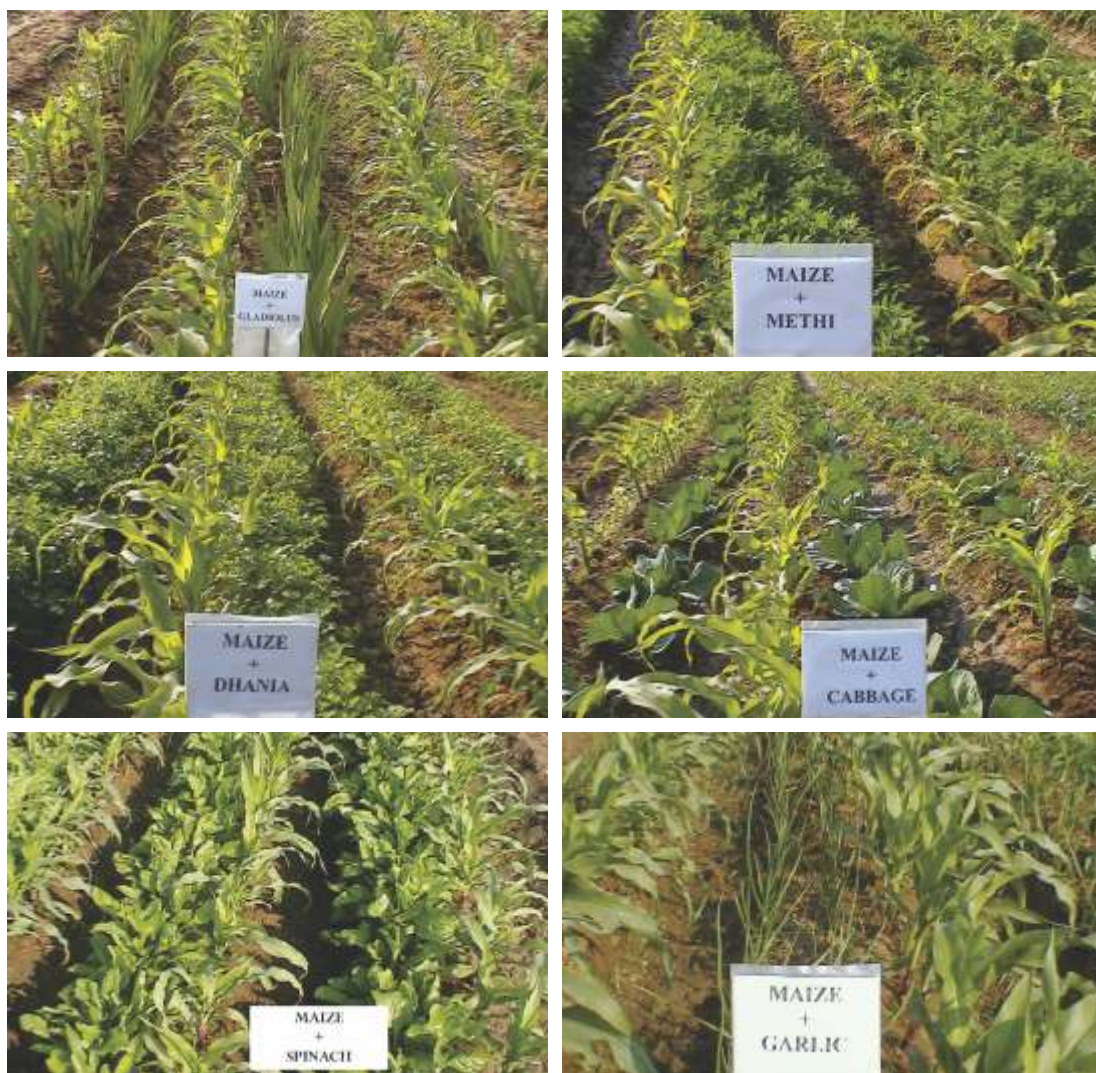


Fig: Maize based intercropping

Fertilizer scheduling: Since baby corn is consumed as vegetable organically grown baby corn has market demand both at local and international level. The organic input mentioned for the normal *kharif* maize to be used for organic baby corn production. For higher yield integrated nutrient management found beneficial for baby corn and organic sources option mentioned in *kharif* maize to be used along with chemical fertilizer mentioned as below:

Crop stage	Schedule	Fertilizer (kg/acre)			
		Urea	DAP	MOP	ZnSO ₄ .7H ₂ O
Before/at sowing	Basal	10	50	40	10
4 leaf stage	1 st split	26	--	--	--
8 leaf stage	2 nd split	40	--	--	--
Before detasseling	3 rd split	32	--	--	--
After detasseling/ one picking	4 th split	20	--	--	--

Irrigation management: Young seedlings, knee high stage, silking and picking are the most sensitive stages for water stress for baby corn and irrigation should be ensured at these stages. Light and frequent irrigations are desirable for baby corn.

Pest Management: As duration of baby corn is short, it experiences lesser extent of pest damage. However, as it is consumed fresh pesticides should be applied judiciously, if required.

Rest management practices for baby corn are similar to normal irrigated maize.

B.3. Sweet corn

This crop gets ready for harvesting just 20 days after pollination i.e. within 65-75 days in *kharif* and 80-100 days in *rabi* season. At high temperature due to conversion of fructose into polysaccharides it losses flavor. Due to this reason, sweet corn cultivation should be avoided in *spring* season. This crop needs isolation of at least 400 m from normal maize as pollination deteriorates its quality. Generally, 10-15 days isolation in sowing time also ensures quality sweet corn. The cobs should be plucked during morning or evening time. Harvested green cobs should be immediately transported to the cold storage in refrigerated trucks. The maize plant parts after harvesting of cobs can be utilized as excellent green fodder. Intercrops as discussed in baby corn can also be grown with sweet corn successfully for higher profitability.

Sowing time: The sowing to be avoided in areas receiving temperature of less than 10C at any of the crop growth stage. At sowing, temperature below 15C may lead to rotten seed. The increase in temperature to more than 35C at tasseling i.e. 45-55 days after sowing leads to tassel blast and lesser grain setting. Therefore, sweet corn can be sown from June to October and in February in Northern India and throughout the year in southern India.

Seed and spacing requirement

Seed rate (kg/acre)	Spacing (cm)	Plant population (per acre)
2.5-3	75x25-30	17,986 to 21,583

Transplanting: Due to costlier seed and lesser plant population, seedling transplanting is beneficial in sweet corn. The seedling should be transplanted when plants are 15-25 cm tall. Transplanting is also beneficial for getting early sweet corn in market for fetching higher prices. However, there should be adequate moisture at the time of transplanting. To avoid damage by ants and other insects the nursery may be treated with furadon or similar granules.

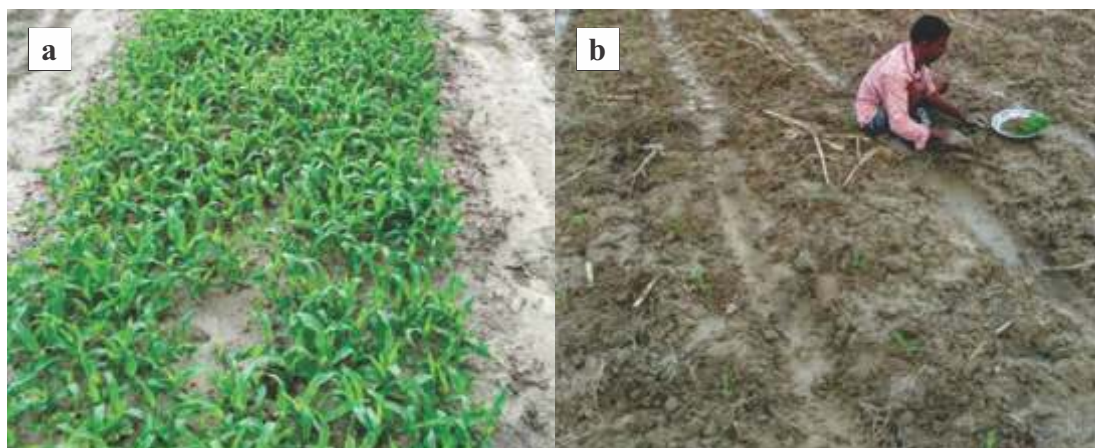


Fig: (a) Maize nursery (b) Transplanting of maize seedlings

Fertilizer scheduling: The organic input mentioned for the normal *kharif* maize can be used for organic sweet corn production. For higher yield, integrated nutrient management found beneficial and organic sources option mentioned in *kharif* maize to be used along with chemical fertilizer mentioned as below:

Crop stage	Schedule	Fertilizer (kg/acre)			
		Urea	DAP	MOP	ZnSO ₄ .7H ₂ O
Before/at sowing	Basal	45	45	30	10
25 DAS	1 st split	35	--	--	--
At flowering	2 nd split	20	--	--	--

Irrigation management: The most critical period to have adequate moisture is during tasseling and silking. Growers should be prepared to irrigate at least 2.5-4.0 cm a week in order to produce high quality sweet corn.

Pest Management: As duration of sweet corn is short and consumed, fresh pesticides should be applied judiciously, if required.

Rest management practices for sweet corn are similar to normal irrigated maize.

B.4. Maize for green cob

The cultivation practices are very similar to sweet corn. The cob plucking should be done at 25-35 days after flowering, when cob colour is still green and kernel are at soft dough stage. The green plants are excellent source of fodder. Remaining practices are same as of sweet corn.

C. Maize as fodder

C.1. Maize for green fodder

Tall, leafy, succulent, late maturing and stay-green type cultivars are desirable for fodder purpose. Sowing should be done using normal seed drill. Weeding is generally not required as the high-density planting leads to less penetration of sunlight resulting in poor growth of weeds.

Sowing time: The sowing to be avoided in areas receiving temperature of less than 10°C at any of the crop growth stages for higher yield of fodder maize. It can be sown round the year in India except second fortnight of December to end of January in Northern India. However, fodder maize is most profitable in *kharif* season, which can be used for silage preparation.

Seed and spacing requirement

Seed rate (kg/acre)	Spacing (cm)	Plant population (per acre)
25	30*10	134,898

Fertilizer scheduling

Crop stage	Schedule	Fertilizer (kg/acre)			
		Urea	DAP	MOP	ZnSO ₄ .7H ₂ O
Before/at sowing	Basal	55	55	30	10
25-30 DAS	1 st split	55	--	--	--

Harvesting should be done just after flowering (at 60-70 DAS) for better quality fodder. Rest agronomic management practices are similar to *kharif* irrigated maize.

C.2. Maize for silage making

All the cultivation practices are similar to *kharif/spring* sown irrigated maize except harvesting, which should be done 25-30 days after flowering (milky to dough stage of grain). Harvesting of maize for silage purpose may be done using Fodder Auto-chopper Loader Machine. Silage can be prepared easily by silo pack machine. This machine can cut and crash the green fodder, load it in the 50 kg air tight plastic bag and compress it. After this process, the bag can be sealed with thread and can be stored in open place for ensiling (process of silage making). These 50 kg capacity bags are easy to handle and transport.

D. Zero till and conservation agriculture based maize cultivation

Under zero till (ZT) cultivation, planting is to be done in stubble of previous crop without any soil disturbance/tillage operation. This saves diesel, tractor's working time and labor and offer timely sowing of succeeding crops. Seed and fertilizers should be placed in band using zero-till Seed-cum-Fertilizer Planter with furrow opener or happy seeder. Weed management is entirely dependent on chemicals as tillage and hand hoeing is prohibited in ZT maize. Destroy the weed before seeding by pre-plant application (10-15 days prior to seeding) of non-selective herbicides, viz., paraquat @ 0.2 kg a.i. per acre in 600 litre water. Under heavy weed infestation, post-emergence application of selective herbicides (mentioned earlier) should be done.

ZT along with surface residue application and diversified crop rotation is known as conservation agriculture. Surface residue retention improves soil physical, chemical and biological activities, reduces soil erosion & evaporation and also moderate soil temperature. Diversified crop rotation break the disease cycle and also improve soil fertility. Zero till planter (under lesser residue load) and happy seeder (under heavy residue load) can be used for planting in no-till field.

Difference in management practices under conventional and conservation agriculture

Practices	Conventional agriculture	Conservation agriculture
Tillage, sowing and intercultural operation	Soil is manipulated 4-5 times for tillage operation, seeding and intercultural operations.	Soil is disturb to a minimum extent as tillage and inter-cultural operations are completely avoided and sowing is done by opening narrow furrow just to put seed.
Cropping system	Mono-cropping/less efficient rotation	Diversified crop rotations
Residue management	Remove/burn	Keep on soil surface
Weed management	Hand weeding and/or herbicide application	Herbicide application

Effect of conventional and conservation agriculture based practices

Particular	Conventional agriculture	Conservation agriculture
Fuel consumption	More	Lesser
Cost of land preparation	More	Lesser
Sowing of succeeding crop	Delay	Timely
Soil health	Poor	Good
Soil compaction	More	Lesser
Water logging	More	Lesser
Soil erosion	More	Less



Fig: (a) Zero till planting of maize in mungbean residue; (b) Surface residue retention in maize

Recommended maize varieties(released in last 6 years)

Variety	Zone	Yield (t/ha)	Maturity (days)	Season	Year of release
Normal maize					
DMRH 1305	NHZ	6.5	Early	Kharif	2018
DMRH-1301	NEPZ, CWZ	9.9	94-96	Rabi	2018
GAWMH-2	Gujarat	-	Early	Kharif	2018
DMRH-1308	CWZ	9.6	92-94	Rabi	2018
Karimnagar Makka	PZ	7.9	90-95	Rainfed kharif	2017
Pant Shankar Makka-4	Uttarakhand	4.6	82-84	Kharif	2017
Central Maize VL 55	NHZ, PZ	7.6	95-97	Kharif	2017
PMH-10	Punjab	11.7	175-180	Spring	2016
Hema	Karnataka	5.6	120-125	Spring	2016
Pratap Makka-9	Rajasthan	4.8	81-87	Kharif, rabi	2016
Palam Shankar Makka-2	NHZ	9.8	Medium	Kharif	2016
Pratap Hybrid	CWZ	5.6	Early	Kharif	2016
Kharif Maize-3					
Karimnagar makka-1	Telangana	9.5	Medium	Kharif ,rabi	2016
PMH-8	Punjab	8.3	112	Spring	2016
Palam shankar makka-1	CWZ	5.3	Late	Kharif	2015
CoH(M) 10	PZ	7.2	Medium	Kharif	2015
HM-13	NHZ	6.6	Early	Kharif	2015
PMH-6	NEPZ	6.3	Medium	Kharif	2015
CoH(M) 7	NEPZ, PZ	7.8	Late	Kharif	2014
CoH(M) 8	NWPZ, NEPZ, PZ, CWZ	7.1	Medium	Kharif	2014
CoH(M) 9	NEPZ, CWZ	6.4	Medium	Kharif	2014
DHM-121	NEPZ, CWZ	5.4	Medium	Kharif	2014
GH-0727	Karnataka	7.5	Late	Kharif	2014
Vivek Maize Hybrid-47	NHZ	6.9	Early	Kharif	2014
Vivek Maize Hybrid -53	NHZ	6.9	Extra-early	Kharif	2014
Vivek Maize Hybrid -51	CWZ	5.1	Extra early	Kharif	2014

Variety	Zone	Yield (t/ha)	Maturity (days)	Season	Year of release
QPM/EDV					
Pusa HM-8 Improved	PZ	6.3	90-95	Kharif	2017
Pusa HM-9 Improved	NEPZ	5.2	85-90		2017
Kharif					
Pusa HM-4 Improved	NWPZ, NEPZ	6.4	90-95	Kharif	2017
Pusa Vivek QPM-9	NHZ, CWZ	5.6	80-85	Kharif	2017
Improved					
Shaktiman-5	NEPZ	5.5	Medium	Kharif &	2018
		& 8.0		Rabi	
Shalimar QPMH-1	J & K	6	135	Kharif	2019
SWEET CORN					
Shalimar Sweet Corn-1	J & K	-	90-95	Kharif	2019
VL Sweet Corn Hybrid-2	NHZ	-	90-95	Kharif	2019
Central Maize CL Sweet	NHZ, NWPZ	10.8	90-95	Kharif	2016
Corn-1					
BABY CORN					
IMHB1539	NHZ	1.3	Short	Kharif	2018
IMHB1532	NWPZ, CWZ	2.0	Short	Kharif	2018
Vivek Hybrid 27	NHZ, CWZ	2.2	95-97	Kharif	2017
POPCORN					
DMRHP-1402	NEPZ, CWZ	3.9	75-77	Kharif	2018
Shalimar Pop Corn-1	NHZ, NWPZ, NEPZ, PZ	3.9	95-100	Kharif	2017
BPCH-6	NHZ, CWZ, PZ, NEPZ, NWPZ	3.2	Early	Kharif	2015

Notes

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